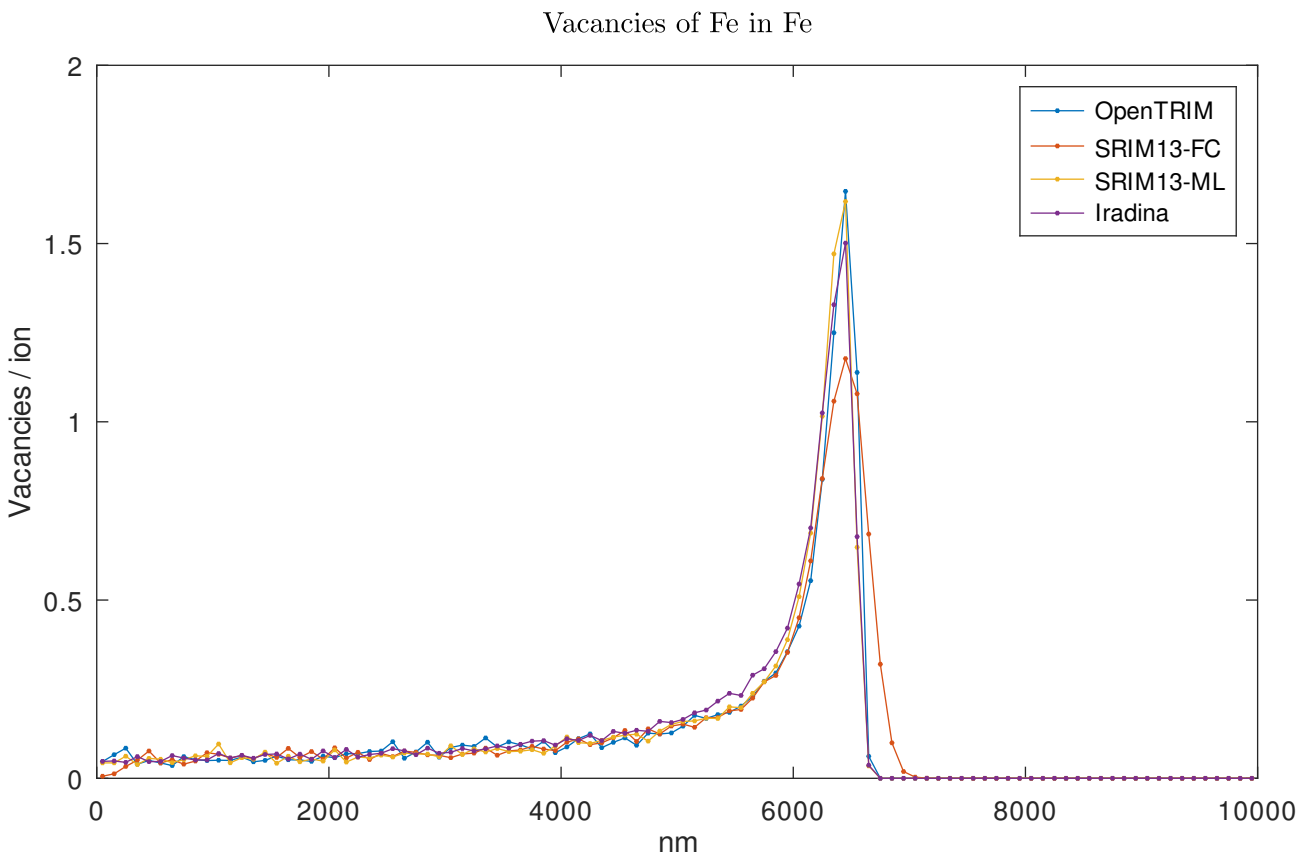
Benchmark #7
1 MeV H on Fe

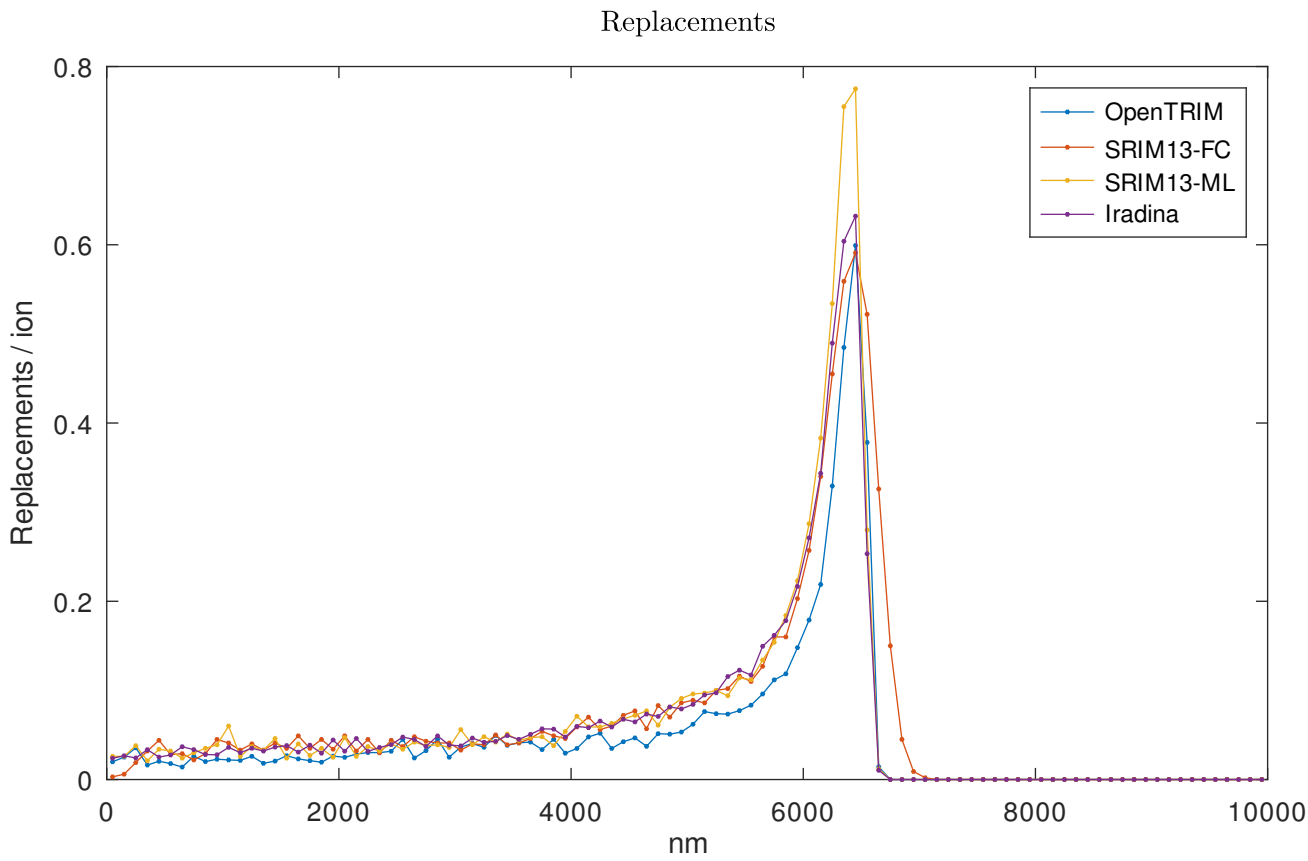
Ion energy E0 = 1e+06 eV

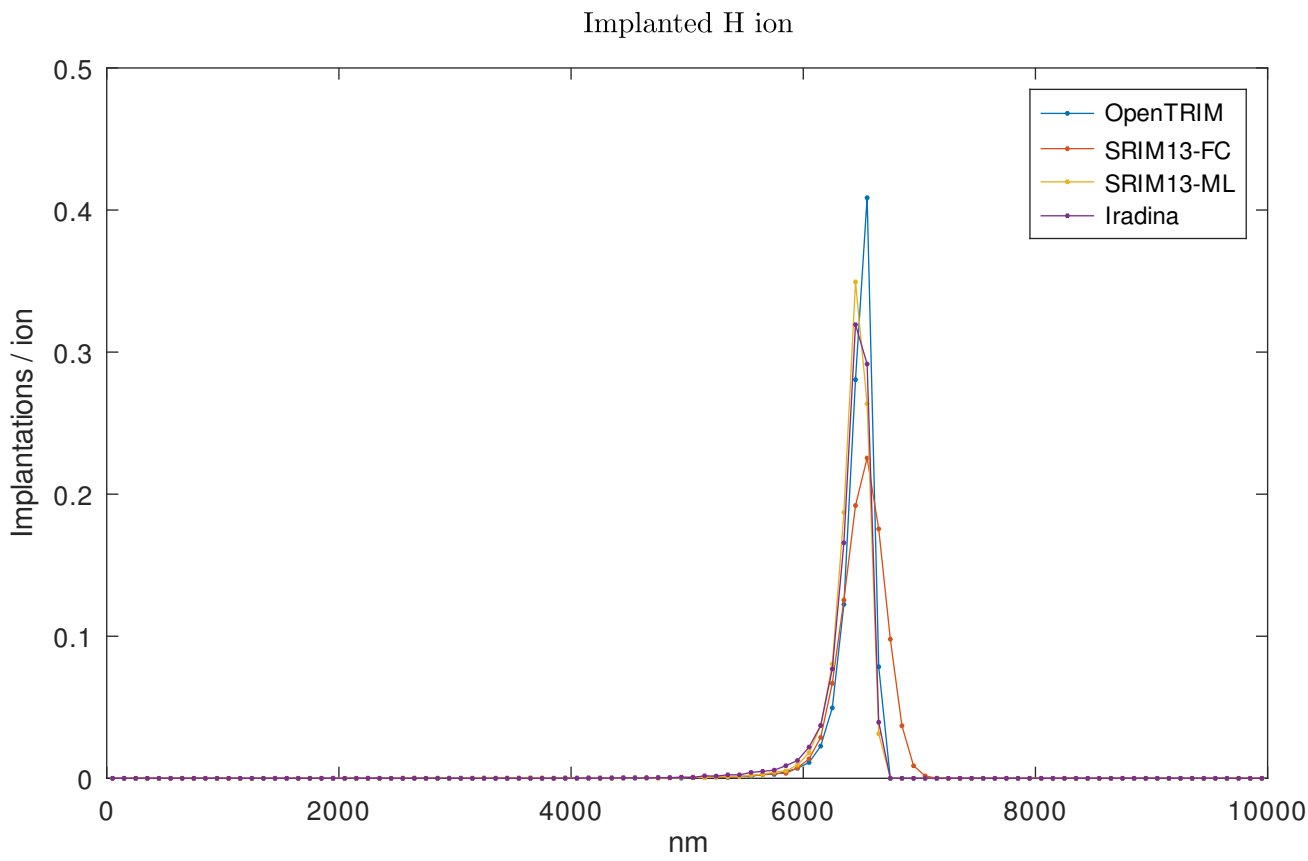
Target depth = 10000 nm

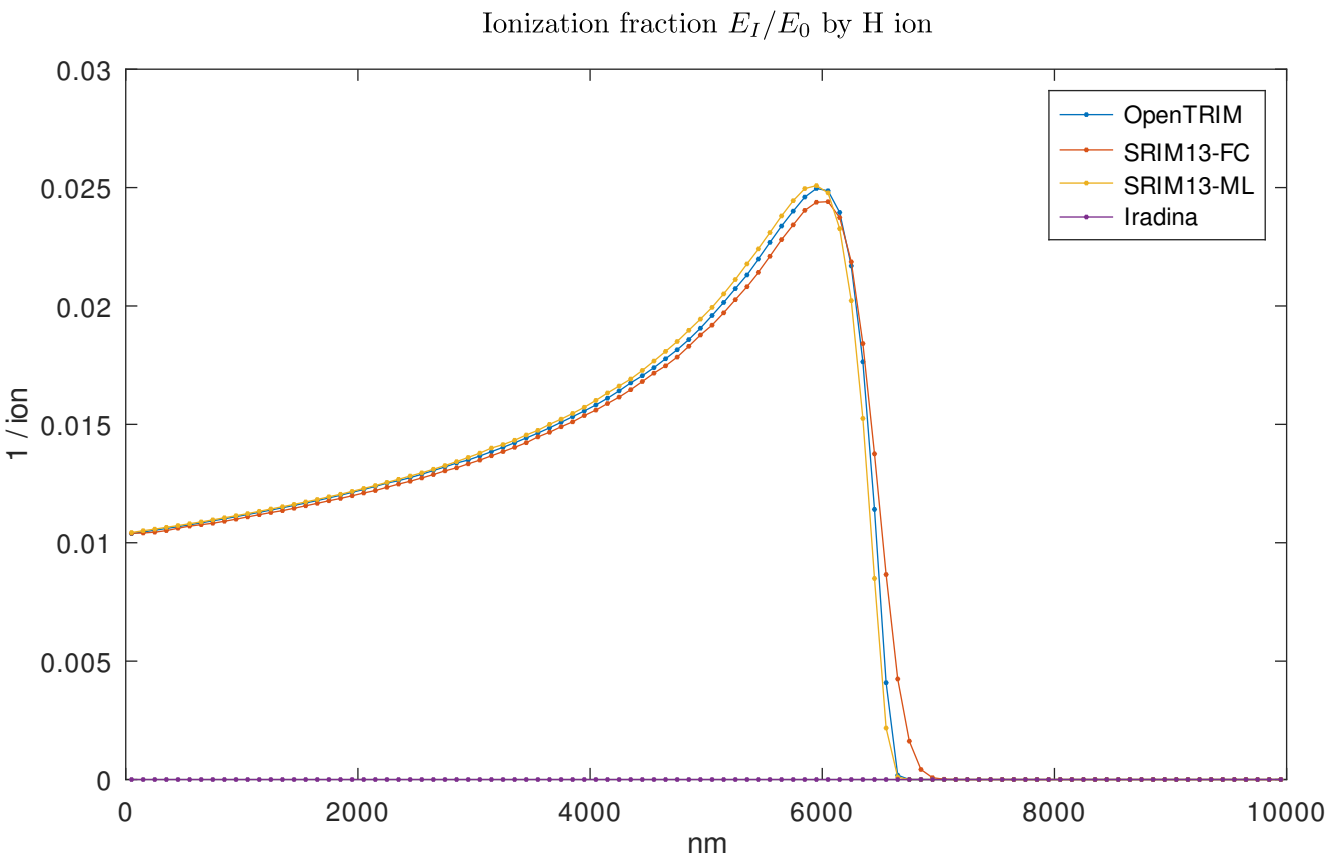
Summary Table

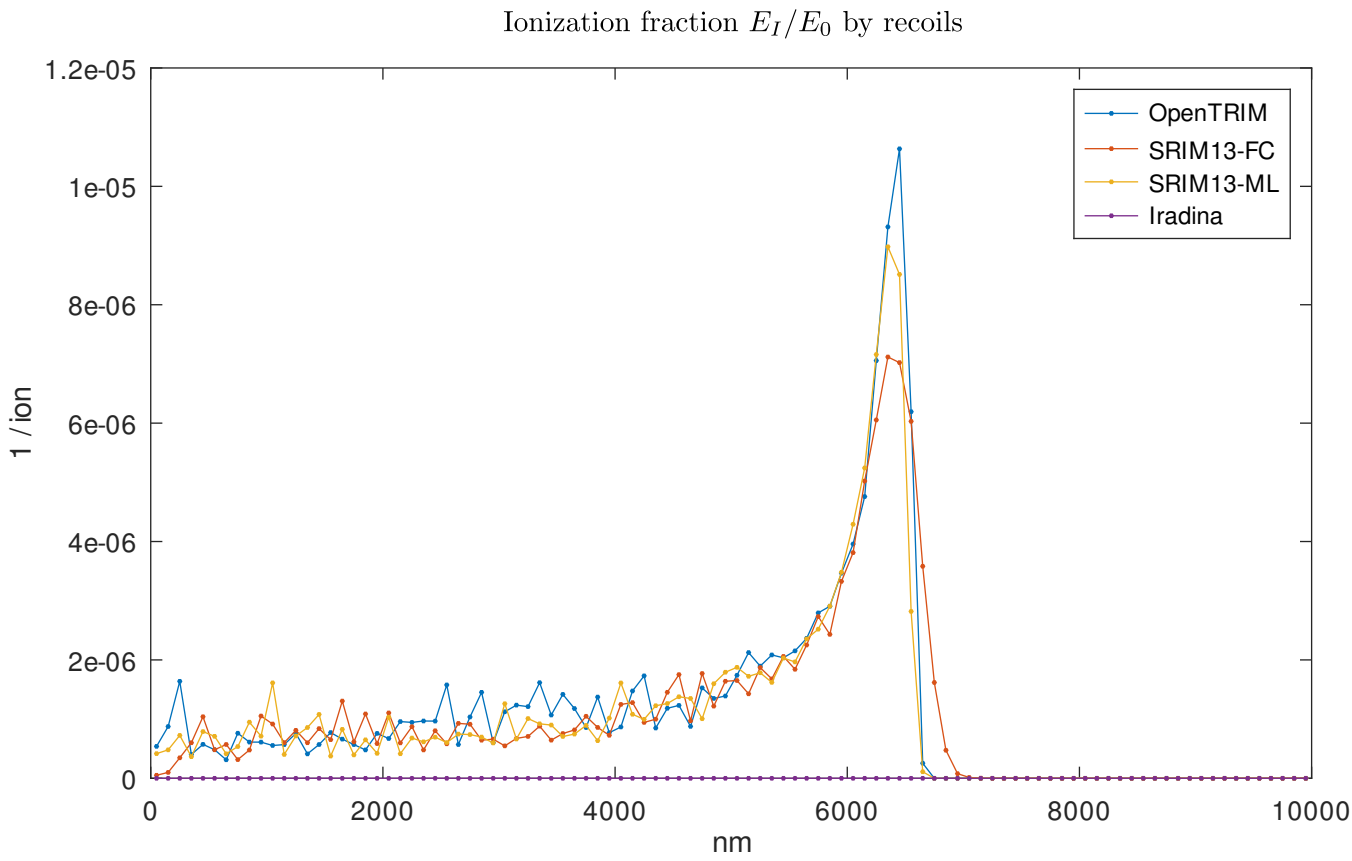
Quantity	OpenTRIM	SRIM13-FC	SRIM13-ML	Iradina
V(Fe)	11.9	12.2	11.9	12.5
R(tot)	4.73	6.69	6.53	6.14
I(H)	1	1	1	1
EI(H)/E0	0.998	0.998	0.998	0
EI(r)/E0	0.000113	0.000104	0.000102	0
EI/E0	0.998	0.998	0.998	0.998
EPh(H)/E0	0.000752	0.000654	0.000915	0
EPh(r)/E0	0.000945	0.00154	0.00178	0
EPh(tot)/E0	0.0017	0.00219	0.00269	0.00183
1-(EI+EPh)/E0	5.96e-08	-0.00055	-0.000704	0

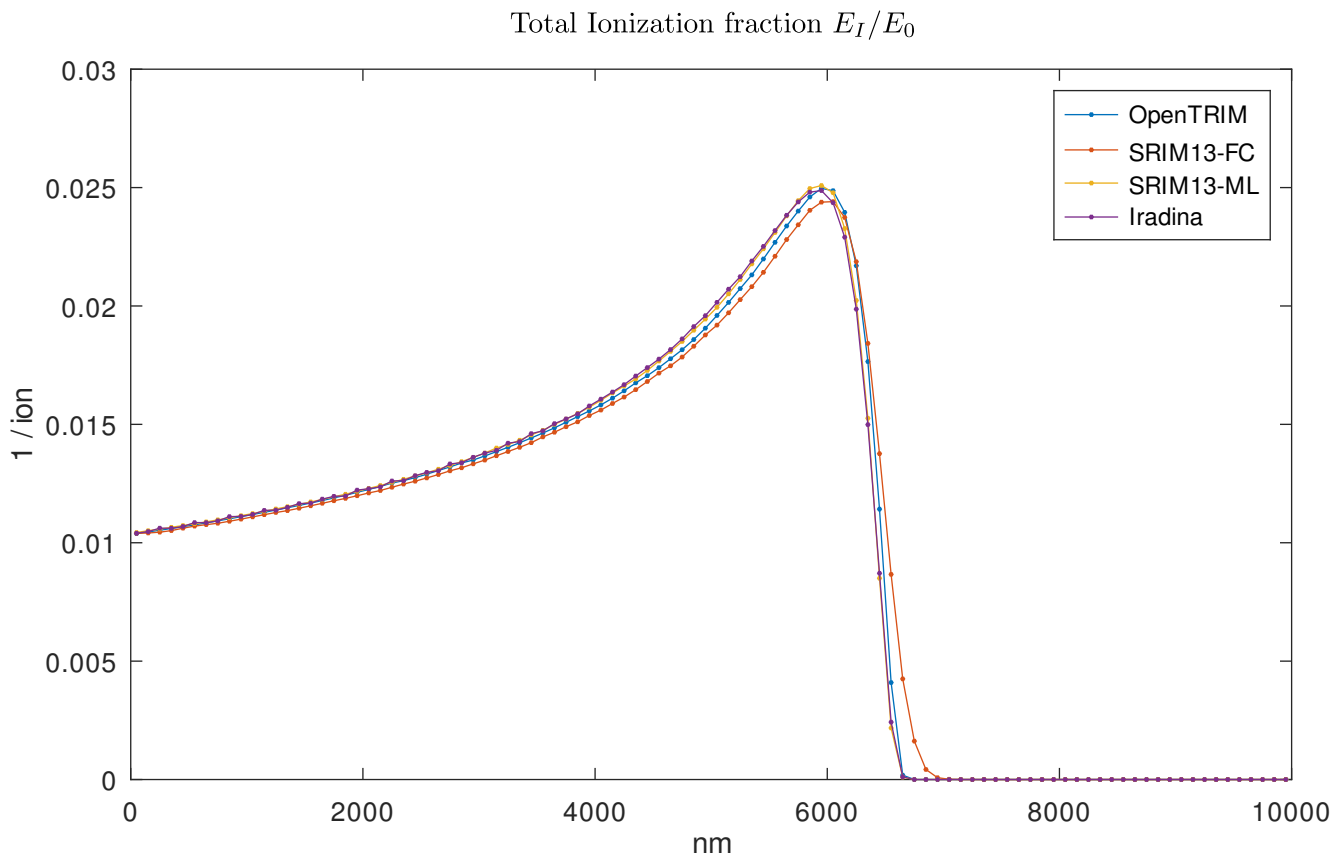

Vacancies of Fe in Fe
Vacancies / ion
nm
OpenTRIM
SRIM13-FC
SRIM13-ML
Iradina
0
0.5
1
1.5
2
0
2000
4000
6000
8000
10000

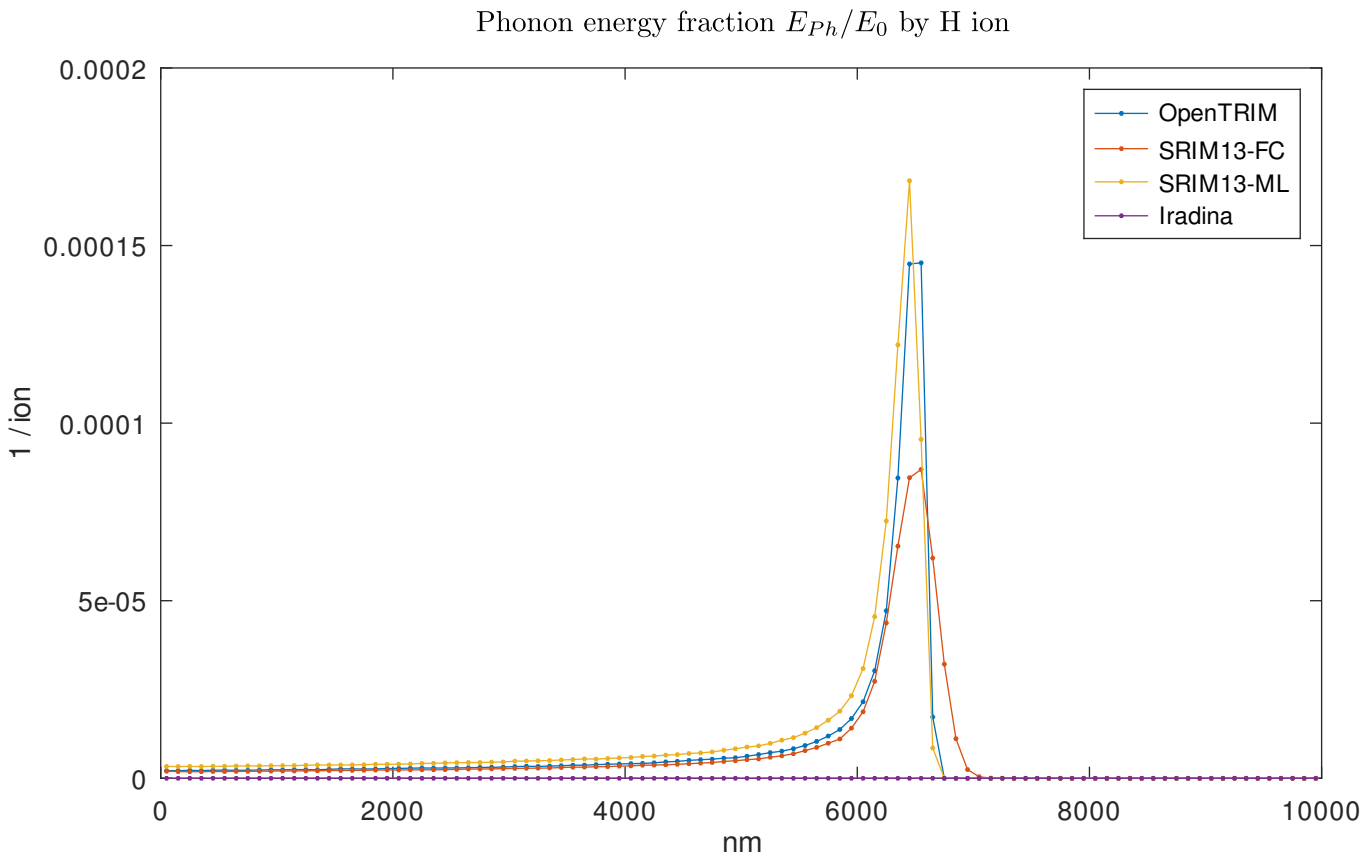

Replacements
Replacements / ion
nm
OpenTRIM
SRIM13-FC
SRIM13-ML
Iradina
0
0.2
0.4
0.6
0.8
0
2000
4000
6000
8000
10000

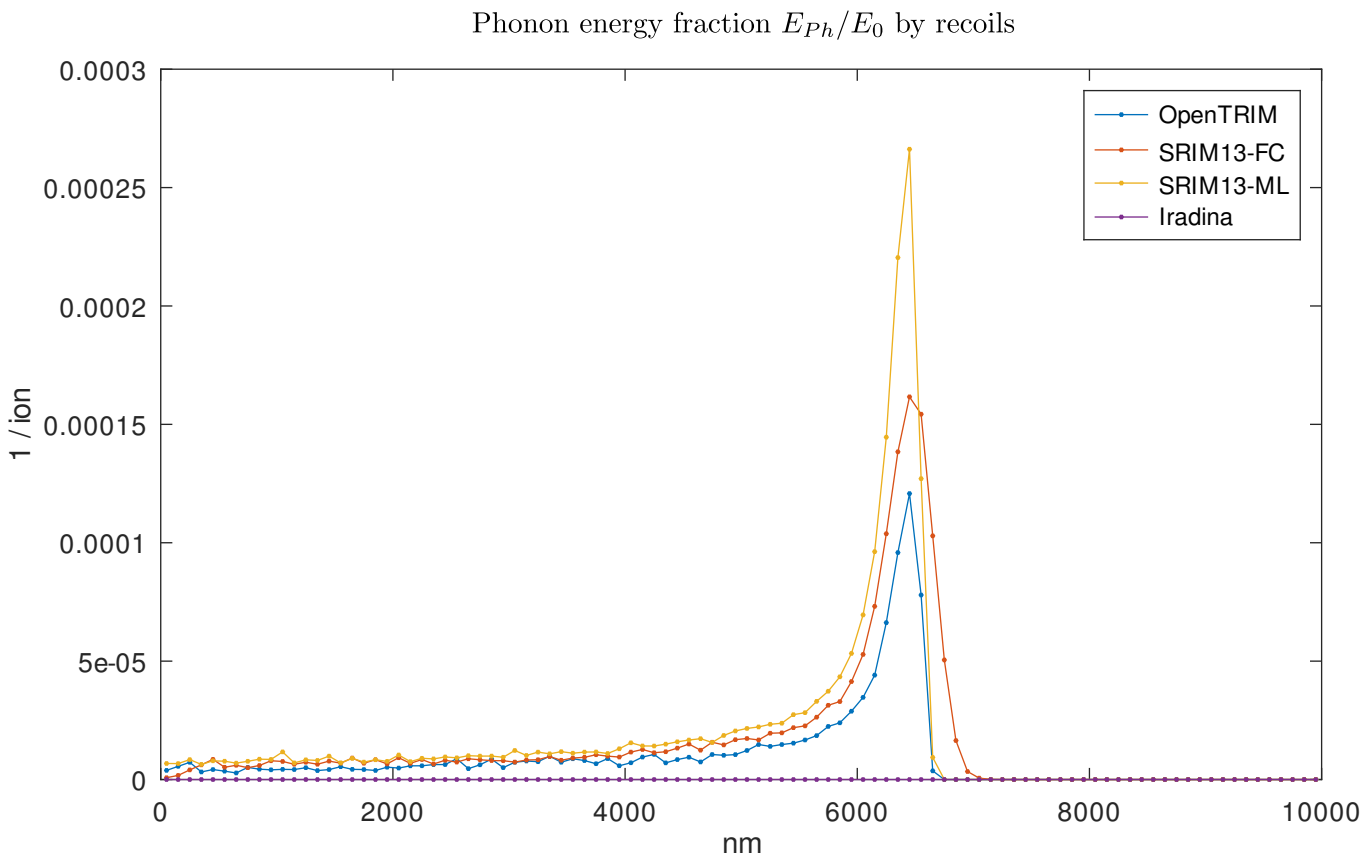

Implanted H ion
Implantations / ion
nm
OpenTRIM
SRIM13-FC
SRIM13-ML
Iradina
0
0.1
0.2
0.3
0.4
0.5
0
2000
4000
6000
8000
10000

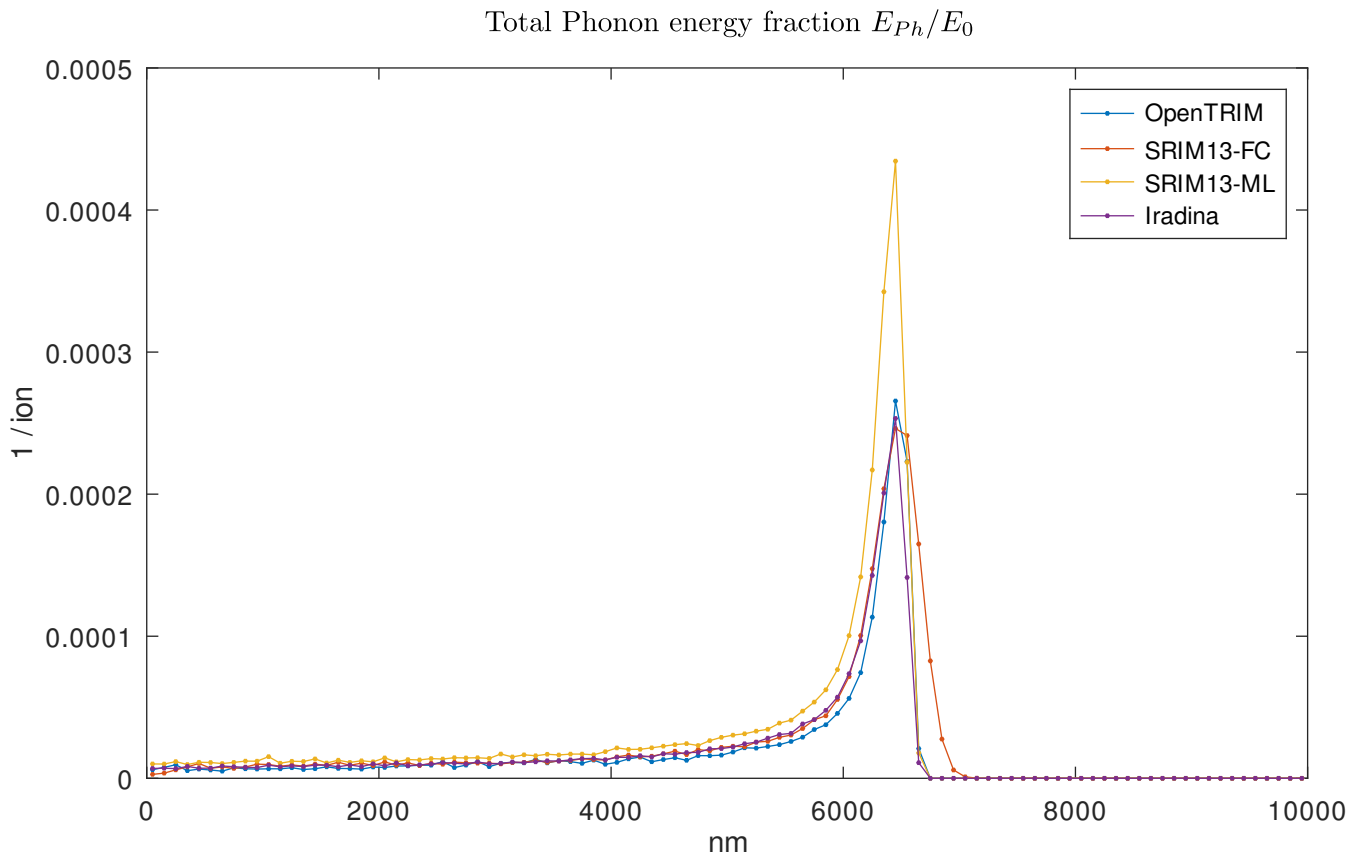

Ionization fraction E_I/E_0 by H ion
1 / ion
nm
OpenTRIM
SRIM13-FC
SRIM13-ML
Iradina
0
0.005
0.01
0.015
0.02
0.025
0.03
0
2000
4000
6000
8000
10000

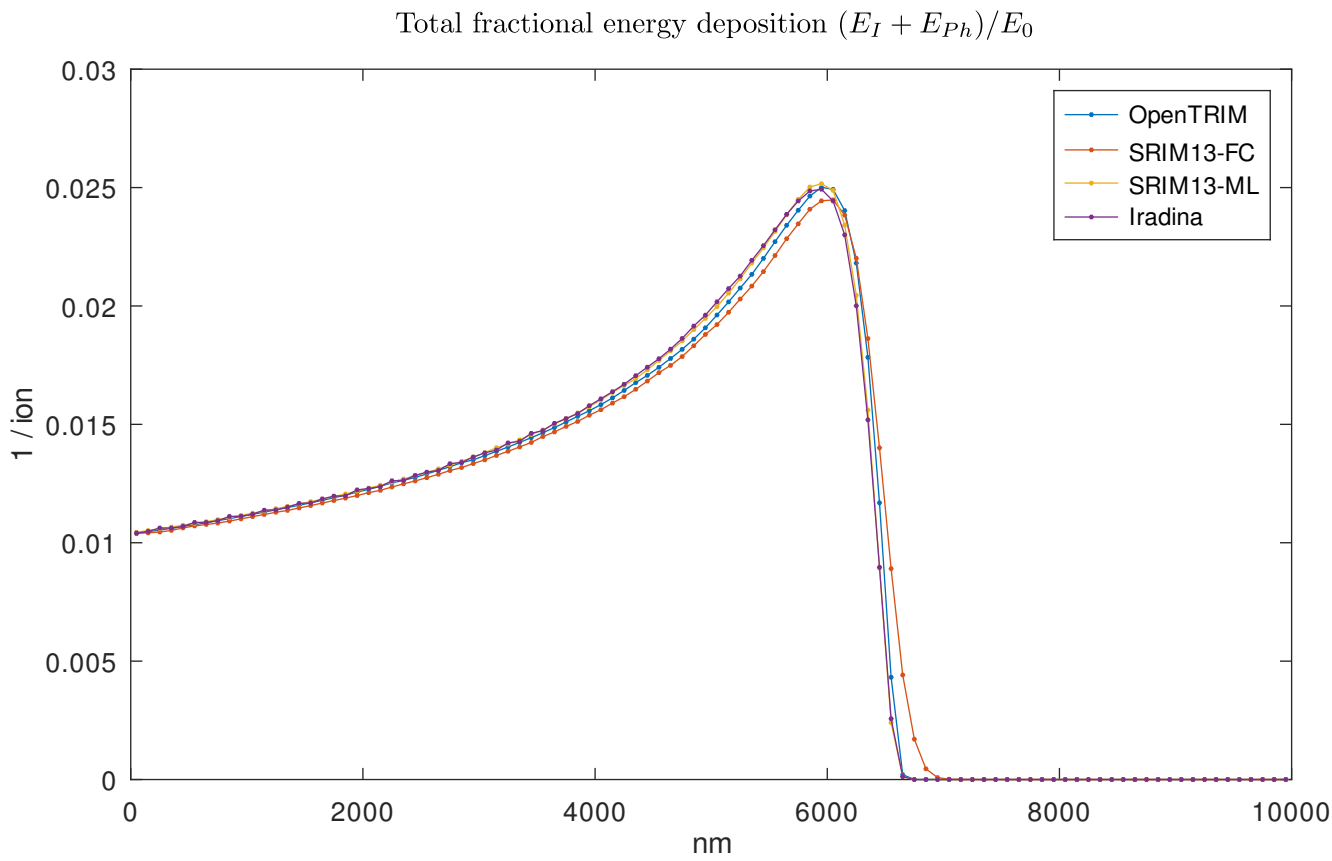

Ionization fraction E_I/E_0 by recoils
1.2e-05
1e-05
8e-06
6e-06
4e-06
2e-06
0
1 / ion
0
2000
4000
6000
8000
10000
nm
OpenTRIM
SRIM13-FC
SRIM13-ML
Iradina


Total Ionization fraction E_I/E_0
0.03
0.025
0.02
0.015
0.01
0.005
0
1 / ion
0
2000
4000
6000
8000
10000
nm
OpenTRIM
SRIM13-FC
SRIM13-ML
Iradina


Phonon energy fraction E_{Ph}/E_0 by H ion
0.0002
0.00015
0.0001
5e-05
0
1 / ion
0
2000
4000
6000
8000
10000
nm
OpenTRIM
SRIM13-FC
SRIM13-ML
Iradina


Phonon energy fraction E_{Ph}/E_0 by recoils
0.0003
0.00025
0.0002
0.00015
0.0001
5e-05
0
1 / ion
0
2000
4000
6000
8000
10000
nm
OpenTRIM
SRIM13-FC
SRIM13-ML
Iradina


Total Phonon energy fraction E_{Ph}/E_0
0.0005
0.0004
0.0003
0.0002
0.0001
0
1 / ion
0
2000
4000
6000
8000
10000
nm
OpenTRIM
SRIM13-FC
SRIM13-ML
Iradina


Total fractional energy deposition $(E_I + E_{Ph})/E_0$
0.03
0.025
0.02
0.015
0.01
0.005
0
1 / ion
0
2000
4000
6000
8000
10000
nm
OpenTRIM
SRIM13-FC
SRIM13-ML
Iradina